

CH 8: NON-METRIC METHODS FOR PATTERN CLASSIFICATION

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Non-Metric Classification

Non-metric classification methods are those that **do not rely on distance measures** (like Euclidean or Manhattan distance) to classify data points.

These methods are useful when **distance is not meaningful** or when the data is **non-numeric, symbolic, or categorical**.

- Focus on **logical rules, decision boundaries, or symbolic processing** rather than distance.
- Often involve **hierarchical, tree-based, or rule-based** structures.
- Well-suited for **nominal/categorical attributes**, such as colors, names, or types.

When to Use:

- When attributes are **qualitative** and cannot be ordered or measured in a conventional sense.
- When **metric-based approaches** (like k-NN, clustering) fail due to non-comparability of feature values.

Examples of Non-Metric Methods:

- Decision Trees (e.g., ID3, C4.5, CART)
- Rule-Based Classifiers
- Symbolic Learning Algorithms
- Some forms of Neural Networks (especially with binary/categorical features)

Advantages:

- Easy to interpret (e.g., decision trees and rule sets)
- Can handle **missing or irrelevant features**
- Naturally support **discrete/nominal data**

Limitations:

- May not generalize well to **continuous or high-dimensional data**
- Prone to **overfitting** (especially in rule-based systems without pruning)

Non-Numeric (Nominal) Data in Classification

Nominal data refers to categorical variables that represent different categories with no inherent order (e.g., gender, color, brand names).

Handling Nominal Data:

- Use symbolic comparison instead of numerical distance.
- Algorithms compare matching categories rather than measure distance.

Techniques for Nominal Data:

- **Decision Trees:** Split nodes based on the presence of specific categorical values.
- **One-Hot Encoding:** Convert nominal attributes into binary features.
- **Rule-Based Learning:** Learn IF-THEN rules that classify based on category membership.
- **Frequency-based Methods:** Use occurrence counts or probability distributions (e.g., Naive Bayes).

Example:

- In metric methods, "Red", "Blue", "Green" cannot be directly compared.
- In non-metric methods (e.g., decision tree), splits can be like:
IF Color = Red THEN Class = Fast

Color	Type	Class
Red	Sports	Fast
Blue	Family	Medium
Green	SUV	Slow

Advantages of Using Non-Metric Methods for Nominal Data:

- Avoids the need for artificial numerical representation.
- Maintains interpretability and clarity.

Challenges:

- Cannot apply arithmetic operations (mean, standard deviation, etc.)
- Cannot use traditional metric-based classifiers unless transformed (e.g., via one-hot encoding)

Encoding Strategies

1. Label Encoding

- Converts each unique category into a **numeric label** (integers).
- **Example:** Color: Red → 0, Blue → 1, Green → 2
- **Pros:** Simple and memory-efficient.
- **Cons:** Implies an order or hierarchy (e.g., $0 < 1 < 2$), which can **mislead algorithms** that assume numerical relationships (like distance-based models).

2. One-Hot Encoding

- Creates a new **binary (0/1) column for each category**.
- **Example:** Color = Red → Color_Red: 1, Color_Blue: 0, Color_Green: 0
- **Pros:** Removes false ordinal relationship; widely used and effective.
- **Cons:** Increases dimensionality—can be **inefficient** for features with **many categories** (high cardinality).

Method	Best For	Caution
Label Encoding	Tree-based or rule-based models	Avoid with distance-based models
One-Hot Encoding	Most ML algorithms	Avoid with high-cardinality data

Aspect	Metric Methods	Non-Metric Methods
Basis	Distance or similarity measures	Logical rules, symbolic representation
Data Type	Mostly numeric	Categorical, nominal, or symbolic
Examples	k-NN, k-Means, SVM (with distance-based kernels)	Decision Trees, Rule-Based Classifiers
Distance Dependency	Strongly dependent on distance metrics	Not dependent on any distance
Handling Categorical Data	Requires encoding (e.g., one-hot)	Handles directly without transformation
Interpretability	Often less interpretable	Usually more interpretable (e.g., tree, rules)
Computational Cost	Can be high for large datasets	Usually efficient in classification
Generalization	Sensitive to data scaling and distribution	Robust to non-numeric and symbolic data
Feature Scaling	Often required (e.g., normalization)	Not needed
Examples of Output	Based on closest data points	Based on learned rules or decision paths

Decision Trees

A Decision Tree is a flowchart-like tree structure where each internal node represents a test on an attribute, each branch represents the outcome of the test, and each leaf node represents a class label (decision).

- **Root Node:** The top node that represents the entire dataset; splits based on the most significant attribute.
- **Internal Nodes:** Represent decisions based on attribute tests.
- **Branches:** Represent the outcome of the test at a node (e.g., yes/no or categorical values).
- **Leaf Nodes (Terminal Nodes):** Represent the final class label or output.
- **Popular algorithms:**
 - ID3 (Iterative Dichotomiser 3)
 - C4.5 / C5.0 (improvement over ID3)
 - CART (Classification and Regression Trees)
- **Pruning (Tree Simplification):** Pruning helps avoid **overfitting** by removing unnecessary branches:
 - **Pre-Pruning:** Stop tree growth early (e.g., minimum samples per node).
 - **Post-Pruning:** Build full tree first, then remove parts using validation data.
- **Handling Different Types of Data**
 - **Categorical Attributes:** Directly split based on category values.
 - **Numerical Attributes:** Use thresholds (e.g., $\text{Age} \leq 30$, $\text{Age} > 30$) to create binary splits.

Advantages of Decision Trees

- **Easy to understand and interpret** (visual flow of decisions).
- Handles both **categorical and numerical data**.
- Requires **little data preprocessing** (no need for normalization).
- Performs **well with non-linear data**.
- **Non-metric:** Doesn't rely on distance functions.

Limitations

- Prone to **overfitting**, especially on noisy data.
- Can be **unstable**: small changes in data can result in different trees.
- Biased towards attributes with **more levels (values)** if using plain information gain.

Splitting Criteria

Used to determine the "**best**" **attribute** to split at each node

a. Information Gain (ID3)

- Measures **reduction in entropy** after the split.
- Higher information gain → better attribute.

b. Gain Ratio (C4.5)

- Adjusts Information Gain by considering **intrinsic information** (to avoid bias toward attributes with many values).

c. Gini Index (CART)

- Measures **impurity** of a node; lower Gini → better split.

Algorithm	Splitting Criteria	Handles Missing Values	Pruning	Output Type
ID3	Information Gain	No	No	Classification
C4.5/C5.0	Gain Ratio	Yes	Yes	Classification
CART	Gini Index	Yes	Yes	Classification & Regression

Example:

A decision tree might start with:

IF Weather = Sunny THEN Play = No

ELSE IF Weather = Overcast THEN Play = Yes

ELSE IF Weather = Rainy THEN ...

Weather	Temperature	Play
Sunny	Hot	No
Overcast	Mild	Yes
Rainy	Cool	Yes

ID3 Algorithm (Iterative Dichotomiser 3)

- Used for generating a decision tree from a set of labeled training data
- Works only with **categorical attributes** (numerical attributes must be discretized).
- Handles only **classification** (not regression).
- Assumes **complete data** (no missing values).

Entropy in Decision Trees

Entropy is a measure of **impurity** or **uncertainty** in a dataset. In the context of decision trees, it tells us how **mixed** the classes are at a particular node.

- If all examples belong to one class, entropy = 0 (pure).
- If examples are evenly distributed among all classes, entropy is maximum.

For a dataset S with classes $C_1, C_2, \dots, C_k$:

- p_i = proportion of samples in class C_i
- $\log_2$ is used to measure in **bits**

$$\text{Entropy}(S) = - \sum_{i=1}^k p_i \log_2 p_i$$

Information Gain in Decision Trees

Information Gain (IG) measures the reduction in entropy after a dataset is split on an attribute. It tells us how much “information” a feature gives us about the class.

- High IG = the attribute **reduces uncertainty** a lot → good for splitting.
 - Low IG = the attribute **adds little value** → not preferred.
- S = current dataset (parent node)
 - A = attribute on which the split is made
 - v = a possible value of attribute A
 - S_v = subset of S where attribute $A = v$
 - $|S|$ = total number of samples
 - $|S_v|$ = number of samples in subset S_v

$$\text{Information Gain}(S, A) = \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} \cdot \text{Entropy}(S_v)$$

Advantages of ID3

- **Simple and intuitive.**
- Builds **interpretable trees.**
- Works well with **categorical data.**

7. Disadvantages

- Can **overfit** if the tree is too deep.
- **Biased** toward attributes with many values.
- Doesn't handle **numeric data** directly.
- No **pruning** mechanism (ID3 grows full tree).

Example:

Age	Competition	Type	Profit (class)
Old	Yes	SW	Down
Old	No	SW	Down
Old	No	HW	Down
Mid	Yes	SW	Down
Mid	Yes	HW	Down
Mid	No	HW	Up
Mid	No	SW	Up
New	Yes	SW	Up
New	No	HW	Up
New	No	SW	Up

- Draw Decision tree using ID3 algorithm
- Find the class label for the data: [Age: Mid, Competition: Yes, Type: HW]

Answer

① Attribute Age

Values(Age) = { Old, Mid, New }

S(Age) = [5 Down, 5 Up]

$$\text{Entropy}(S) = -\frac{5}{10} \log\left(\frac{1}{2}\right) - \frac{5}{10} \log\left(\frac{1}{2}\right) = 1$$

① S_{Old} = [3 D, 0 U]

$$E(S_{\text{Old}}) = -\frac{3}{3} \log(1) - 0 = 0$$

② S_{Mid} = [2 D, 2 U], $E = -\frac{2}{4} \log\left(\frac{1}{2}\right) - \frac{2}{4} \log\left(\frac{1}{2}\right) = 1$

③ S_{New} = [0 D, 3 U], $E = 0$

$$\text{Gain}(S, \text{Age}) = 1 - \left(\frac{3}{10} \times 0 + \frac{4}{10} \times 1 + \frac{0}{10} \times 0\right) = 0.6$$

② Attribute Competition

Value(Com) = { Yes, No }

① S_{Yes} = [3 A, 1 Y], $E = -\frac{3}{4} \log\left(\frac{3}{4}\right) - \frac{1}{4} \log\left(\frac{1}{4}\right)$
 $= +0.31127 + 0.5 = 0.81127$

② S_{No} = [2 D, 4 Y], $E = -\frac{2}{6} \log\left(\frac{1}{3}\right) - \frac{4}{6} \log\left(\frac{2}{3}\right)$
 $= 0.91829$

$$\text{Gain}(S, \text{Comp}) = 1 - \left(\frac{4}{10} \times 0.81127 + \frac{6}{10} \times 0.91829\right)$$

$$= 1 - (0.32451 + 0.55097)$$

$$= 0.12452$$

⑤ Attribute Type

$$V(\text{Type}) = \{SW, HW\}$$

$$S_{SW} = [3D, 3U], \quad E = -\frac{3}{6} \log\left(\frac{1}{2}\right) - \frac{1}{2} \log\left(\frac{1}{2}\right) = 1$$

$$S_{HW} = [2D, 2U], \quad E = -\frac{2}{4} \log\left(\frac{1}{2}\right) - \frac{1}{2} \log\left(\frac{1}{2}\right) = 1$$

$$\text{Gain}(S, \text{Type}) = 1 - (0.6 \times 1 + 0.4 \times 1) = 0$$

As $\text{Gain}(S, \text{Age})$ is highest, Age is root node



Age	Comp	Type	Profit
Mid	yes	SW	Down
Mid	yes	HW	Down
Mid	No	HW	Up
Mid	No	SW	Up

$$\text{Value}(\text{mid}) = [\text{Down}, \text{Up}]$$

$$E(\text{mid}) = \frac{2}{4} \log\left(\frac{1}{2}\right) - \frac{1}{2} \log\left(\frac{1}{2}\right) = 1$$

① Comp

$$\text{Value}(C) = \text{yes, No}$$

$$\text{① } S_y = [2D, 2U], \quad E = \frac{2}{2} \log(1) - 0 = 0$$

$$\text{② } S_n = [2U, 0D], \quad E = 0$$

$$\text{Gain}(S_{\text{mid}}, \text{Comp}) = 1 - 0 = 1$$

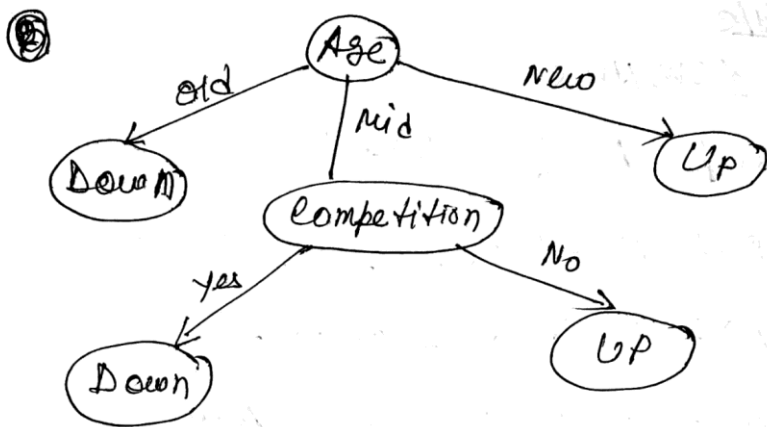
① ② Type

$$\text{Value}(T) = \{SW, HW\}$$

$$\text{① } S_{SW} = [1D, 1U], \quad E = \frac{1}{2} \log\left(\frac{1}{2}\right) - \frac{1}{2} \log\left(\frac{1}{2}\right) = 1$$

$$\text{② } S_{HW} = [1D, 1U], \quad E = 1$$

$$\text{Gain}(S_{\text{mid}}, \text{Type}) = 1 - (0.5 \times 1 + 0.5 \times 1) = 0$$



⑥ Also we need to predict class label for the data
 [Age = mid, Competition = Yes, Type = HW]
 Ans. Class label is Down

CART Algorithm (Classification and Regression Trees)

- Used for constructing **binary decision trees** for **both classification and regression** tasks.
- **Binary Trees Only**: Each internal node splits into exactly two branches (unlike ID3/C4.5 which can have multiple).
- Can handle both:
 - **Classification problems** (discrete output)
 - **Regression problems** (continuous output)
- Supports both **numerical** and **categorical** attributes.
- Uses **Gini Index** (for classification) instead of Entropy.
- **Split the dataset** recursively using a feature and a threshold that results in the **most “pure”** subsets.
- Use different **impurity measures** for classification and regression.

Handling Data Types

- **Numerical Features**: CART finds the best **threshold** (e.g., Age < 35).
- **Categorical Features**: Handles binary split (e.g., Type = HW or not).
- **Missing Values**: Can handle using surrogate splits.

Impurity Measures

A. For Classification: Gini Index

$$\text{Gini}(D) = 1 - \sum_{i=1}^C p_i^2$$

- p_i = proportion of class i in dataset D
- Lower Gini = purer node

B. For Regression: Mean Squared Error (MSE)

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y})^2$$

- y_i = true value,
- $\hat{y}$ = predicted mean value

7. Advantages of CART

- Supports **both classification and regression**
- Handles **numerical and categorical data**
- Creates **binary splits**, making the tree more consistent
- Includes **pruning** (post-pruning with cost-complexity pruning)
- Often more **robust** than ID3/C4.5

8. Disadvantages

- Can still **overfit** without pruning
- Binary splits may lead to **deeper trees**
- Sensitive to **small data changes** (can change structure)
- May be **unstable** (small data changes can lead to different trees)
- Biased toward features with **more split points**
- Trees may become **complex and hard to interpret** if not constrained

Example

CGPA	Interactive	Practical Knowledge	Comm Skills	Job Offer
>=9	Yes	Very good	Good	Yes
>=8	No	Good	Moderate	Yes
>=9	No	Average	Poor	No
<8	No	Average	Good	No
>=8	Yes	Good	Moderate	Yes
>=9	Yes	Good	Moderate	Yes
<8	Yes	Good	Poor	No
>=9	No	Very good	Good	Yes
>=8	Yes	Good	Good	Yes
>=8	Yes	Average	Good	Yes

Answer

$$\boxed{\text{Gini Index} = 1 - \sum_{i=1}^m p_i^2}$$

$$\text{Gini(Table)} = 1 - \left(\frac{7}{10}\right)^2 - \left(\frac{3}{10}\right)^2 = 1 - 0.49 - 0.09 \\ = 0.42$$

① Attribute CGPA

Values of CGPA: $\geq 9, \geq 8, < 8$



CGPA	Yes	No
≥ 9	3	1
≥ 8	4	0
< 8	0	2

We need to find Gini-index
of each subset of CGPA
3 values of CGPA
 $\therefore 2^3$ subsets.

Subsets: $\{1\}, \{1 \geq 9\}, \{1 \geq 8\}, \{1 < 8\}, \{1 \geq 9, \geq 8\}, \{1 \geq 9, < 8\},$
 $\{1 \geq 8, < 8\}, \{1 \geq 9, \geq 8, < 8\}$

$$GI(T, \{1 \geq 9, \geq 8\}) = 1 - \left(\frac{3+4}{8}\right)^2 - \left(\frac{1}{8}\right)^2 = 1 - \frac{49}{64} - \frac{1}{64} \\ = \frac{64 - 49 - 1}{64} = \frac{14}{64} = 0.21875$$

$$GI(T, \{1 < 8\}) = 1 - \left(\frac{0}{2}\right)^2 - \left(\frac{2}{2}\right)^2 = 0$$

$$GI(T, \{1 \geq 9, \geq 8, < 8\}) = \frac{8}{10} \times 0.21875 + \frac{2}{10} \times 0 \\ = 0.175$$

$$GI(T, \{1 \geq 9, < 8\}) = 1 - \left(\frac{3}{6}\right)^2 - \left(\frac{3}{6}\right)^2 = 0.5$$

$$GI(T, \{1 \geq 8\}) = 1 - \left(\frac{4}{4}\right)^2 - \left(\frac{0}{4}\right)^2 = 0$$

$$GI(T, \{1 \geq 9, < 8, \geq 8\}) = \frac{6}{10} \times 0.5 + \frac{4}{10} \times 0 = 0.3$$

$$GI(T, \{1 \geq 8, < 8\}) = 1 - \left(\frac{4}{6}\right)^2 - \left(\frac{2}{6}\right)^2 = 0.4444$$

$$GI(T, \{1 \geq 9\}) = 1 - \left(\frac{3}{4}\right)^2 - \left(\frac{1}{4}\right)^2 = 0.375$$

$$GI(T, \{1 \geq 9, < 8, \geq 8\}) = \frac{6}{10} \times 0.4444 + \frac{4}{10} \times 0.375 \\ = 0.41664$$

Now,

Subsets	G.I
$\{ \geq 9, \geq 8 \}, < 8 \}$	0.175
$\{ \geq 9, < 8 \}, \geq 8 \}$	0.3
$\{ < 8, \geq 8 \}, \geq 9 \}$	0.41667

we choose min.
G.I for best splitting.

$\therefore$ CGPA = $\{ \geq 9, \geq 8 \}, < 8 \}$ is selected.

Now $\Delta \text{Gini}(\text{CGPA}) = \text{Gini}(T) - \text{Gini}(T, \text{CGPA})$
 $= 0.42 - 0.175 = \boxed{0.2445}$

② Attribute: Int.

Int	Yes	No
Yes	5	1
No	2	2

Subsets: {Yes, No}

$\text{G.I}(T, \text{Yes}) = 1 - \left(\frac{5}{6}\right)^2 - \left(\frac{1}{6}\right)^2 = 0.278$

$\text{G.I}(T, \text{No}) = 1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 = 0.5$

$\therefore \text{G.I}(T, \{Y, N\}) = \frac{6}{10} \times 0.278 + \frac{4}{10} \times 0.5$
 $= 0.3668$

$\Delta \text{Gini}(\text{Int}) = 0.42 - 0.3668 = \boxed{0.0532}$

③ $\Delta \text{Gini}(\text{prec}) = \boxed{0.1146}$

④ $\Delta \text{Gini}(\text{comm}) = \boxed{0.2445}$

⑤ $\Delta \text{Gini}(\text{CGPA}) = \boxed{0.2445}$

we can choose
between (any one)

lets choose CGPA.



CGPA	Int	prec.	comm.	Sob.
≥ 9	Y	very good	good	Y
≥ 8	N	good	moderate	Y
≥ 9	N	avg	bad	N
≥ 8	Y	good	mod.	Y
≥ 9	Y	good	mod	Y
≥ 8	N	very good	good	Y
≥ 8	Y	good	good	Y
≥ 8	Y	avg	good	Y

$$Gini(T) = 1 - \left(\frac{7}{8}\right)^2 - \left(\frac{1}{8}\right)^2 = 0.21875$$

① Int

Int	Yes	No
Y	5	0
N	2	1

$$Gini(T, Yes) = \frac{1}{8} - \left(\frac{5}{8}\right)^2 - \left(\frac{0}{8}\right)^2 = 0$$

$$Gini(T, N) = 1 - \left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)^2 = 0.444$$

$$Gini(T, (Y, N)) = \left(\frac{5}{8}\right)^2 \times 0 + \left(\frac{3}{8}\right)^2 \times 0.444 = 0.06244$$

$$\Delta Gini(T) = 0.21875 - 0.06244 = 0.15631$$

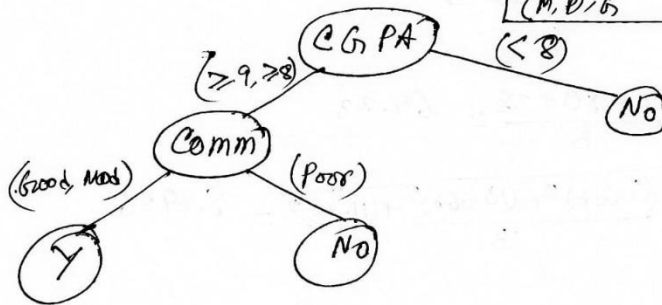
② Proc

$$\Delta Gini(P) = 0.0934$$

③ Comm

$$\Delta Gini(C) = 0.2184$$

(V6, 6) A	0.125
(V6, A, 6)	0.1875
(A, 6), V6	0.2085
(G, M) P	0
(G, P), M	0.2
(M, P), G	0.1875



Aspect	Gini Index	Entropy
Definition	Measures the impurity of a dataset as the probability of misclassification.	Measures the impurity based on information theory — amount of information needed to classify.
Used In	Used by CART algorithm.	Used by ID3 and C4.5 algorithms.
Formula	$Gini(D) = 1 - \sum_{i=1}^C p_i^2$	$Entropy(S) = - \sum_{i=1}^k p_i \log_2 p_i$
Range	0 (pure) to < 0.5 (for 2 classes)	0 (pure) to $\log_2(C)$, C = number of classes
Computation	Faster and simpler to calculate.	Slightly more computationally expensive.
Split Bias	Prefers splits with larger class dominance .	Prefers balanced splits (more sensitive to class distribution).
Units	No specific units.	Measured in bits (information gain).

Aspect	CART	ID3
Definition	Decision Tree algorithm that uses Gini Index and supports classification and regression.	Decision Tree algorithm that uses Entropy and Information Gain for classification only.
Tree Type	Always creates binary splits (2 branches).	Creates multi-way splits (branches = number of attribute values).
Splitting Criterion	Uses Gini Index .	Uses Information Gain (Entropy) .
Supports Regression	Yes .	No , only classification.
Numerical Attributes	Directly supported.	Not directly supported (need discretization).
Pruning	Includes post-pruning (cost-complexity pruning).	No pruning (builds full tree).
Output Tree Depth	Can be deeper due to binary splits.	Usually shallower with multi-way splits.
Overfitting	Handles it better with pruning.	Prone to overfitting .

Regression Tree (type of decision tree)

A **Regression Tree** is a type of **Decision Tree** used for **predicting continuous (real-valued) outputs** instead of categorical labels.

- Unlike classification trees (which predict classes like *Yes/No*), regression trees predict values like *salary, temperature, profit, etc.*
- Goal:** To partition the input space into regions where the **target variable's variance is minimized** within each region, and the **mean value** of the region is used for prediction.

Example

S.No.	Assessment	Assignment	Project	Result (%)
1	Good	Yes	Yes	95
2	Average	Yes	No	70
3	Good	No	Yes	75
4	Poor	No	No	45
5	Good	Yes	Yes	98
6	Average	No	Yes	80
7	Good	No	No	75
8	Poor	Yes	Yes	65
9	Average	No	No	58
10	Good	Yes	Yes	89

Answer

$$\text{Mean} = \frac{95 + 70 + 75 + 45 + 98 + 80 + 75 + 65 + 58 + 89}{10} = 75$$

$$Sd = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N}} = \sqrt{\frac{(20)^2 + 5^2 + 0^2 + 20^2 + 23^2 + 5^2 + 0^2 + 10^2 + 17^2 + 14^2}{10}} = 15.6971$$

1) Assessment values / good, Avg, Poor

(i) Good

$$\text{Mean} = \frac{95 + 75 + 98 + 75 + 89}{5} = 86.4$$

$$Sd = \sqrt{\frac{(8.6)^2 + (11.4)^2 + (11.6)^2 + (11.4)^2 + (2.6)^2}{5}} = 9.2488$$

(ii) Avg

$$\text{Mean} = \frac{70 + 80 + 58}{3} = 69.33$$

$$Sd = \sqrt{\frac{(0.66)^2 + (10.66)^2 + (11.33)^2}{3}} = 8.9926$$

(iii) Poor

$$\text{Mean} = \frac{45 + 65}{2} = 55$$

$$Sd = \sqrt{\frac{10^2 + 10^2}{2}} = 10$$

$$\Delta Sd = 15.6971 - \left(\frac{5}{10} \times 9.2488 + \frac{3}{10} \times 8.9926 + \frac{2}{10} \times 10 \right)$$

$$= 15.6971 - 9.57218 = \boxed{6.12492}$$

2) Assignment

values (Yes, No)

(i) Yes

$$\text{Mean} = \frac{95 + 70 + 98 + 65 + 89}{5} = 83.4$$

$$Sd = \sqrt{\frac{(11.6)^2 + (13.4)^2 + (14.6)^2 + (18.4)^2 + (4.6)^2}{5}} = 13.3192$$

(ii) No

$$\text{Mean} = \frac{75 + 45 + 80 + 75 + 58}{5} = 66.6$$

$$Sd = \sqrt{\frac{(8.4)^2 + (21.6)^2 + (13.4)^2 + (8.4)^2 + (8.6)^2}{5}} = 13.1240$$

$$\Delta Sd = 15.6971 - \left(\frac{3}{5} \times 13.3192 + \frac{2}{5} \times 13.1240 \right)$$

$$= \boxed{2.4755}$$

3) Prf.

$$\text{① Sd of Yes} = 12.6 \quad \Delta Sd = \boxed{3.63}$$

$$\text{② Sd of No} = 13.39$$

We chose largest S.d. which is 6.1249.

